



DashGrab : Shop Swiftly

F24-221-D-DashGrab

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Contents

1	Introduction	2
1.1	Problem Statement	2
1.2	Scope	2
1.3	Solution	2
2	Software Requirements Specification (SRS)	3
2.1	Functional Requirements	3
2.2	Non-Functional Requirements	3
2.3	User Classes and Characteristics	3
3	Software Design	5
3.1	Design Methodology	5
3.2	Architectural Style	5
4	UML Diagrams	6
4.1	Architectural Diagram	7
4.2	Activity Diagram	8
4.3	Entity-Relationship Diagram (ERD)	8
5	Validation and Testing	9
5.1	Testing Approach	9
5.2	Results	9
6	Conclusion	10

Chapter 1

Introduction

DashGrab is an innovative project designed to automate the shopping experience in retail clothing stores by combining Computer Vision, RFID technology, and automated billing. The goal is to streamline the shopping process, making it more efficient and secure.

1.1 Problem Statement

Traditional retail systems require manual checkout processes or expensive infrastructure upgrades to automate purchasing, leading to inefficiencies and long wait times. These methods often fail to ensure real-time tracking of customer activities or detect unauthorized actions, resulting in potential security risks. Additionally, implementing automated systems with dedicated smart cameras can be costly. There is a need for a system that can use existing CCTV setups to ensure seamless purchasing, real-time activity tracking, and automated billing.

1.2 Scope

This project aims to develop a system that integrates with current CCTV infrastructure to track customers' shopping behavior in a retail environment, specifically targeting clothing stores. The system will recognize individuals, track their activities (like picking up or returning clothing items), and automatically update their accounts with purchases. The solution will include real-time detection and monitoring to ensure accurate billing and security checks, offering a balance between convenience and operational efficiency.

1.3 Solution

DashGrab uses Computer Vision to monitor customer behavior, tracking their actions in real-time, such as picking up or returning items. It integrates RFID tags for seamless item tracking, eliminating manual checkouts by associating products with the customer automatically. The system also features automated billing, charging customers as they exit the store for a frictionless experience. Additionally, DashGrab includes security measures to detect shoplifting and fraud, identifying suspicious activities like unauthorized item removal. This combination of technology improves both shopping convenience and security, streamlining retail operations efficiently.

Chapter 2

Software Requirements Specification (SRS)

2.1 Functional Requirements

The following are the core functionalities of the DashGrab system:

- **User Identification:** The system identifies customers upon entry using camera surveillance and registered profiles.
- **Item Detection:** RFID tags attached to clothing items are detected automatically as the customer moves within the store.
- **Automated Checkout:** Once the customer exits the store, the system automatically processes payment through the registered payment method.
- **Security Monitoring:** Continuous monitoring for suspicious activity, such as shoplifting or insufficient balance during checkout.

2.2 Non-Functional Requirements

Non-functional aspects include:

- **Real-Time Processing:** The system must process transactions and monitor security in real-time to ensure smooth shopping.
- **Security:** The system should detect and prevent fraud, including monitoring for attempts to steal items or checkout with insufficient funds.
- **Availability:** The system should have high availability and minimal downtime to support continuous operations.

2.3 User Classes and Characteristics

This section identifies the various user roles that will interact with the Dashgrab system.

Users	Description
Customer	The main users of the system. Customers are identified upon entering the store, and their actions, such as picking or returning items, are tracked. Each customer has an account that will be billed based on their shopping.
Staff	Responsible for managing the Dashgrab system, ensuring accurate detection of customers and items. They can also review and resolve billing or detection issues.
Admin	Administrators oversee the system's operation, monitoring performance, and security. They manage user permissions and ensure data integrity.

Table 2.1: User Classes and Characteristics for DashGrab

Chapter 3

Software Design

3.1 Design Methodology

DashGrab employs an Object-Oriented Programming (OOP) methodology to handle the modular nature of the system. This design approach allows for easy scalability, code reusability, and efficient management of different modules such as customer identification, item detection, and billing automation.

3.2 Architectural Style

The architectural style chosen for DashGrab is Microservices Architecture. Each module, including item detection, billing, and security monitoring, functions independently but communicates with the overall system for real-time synchronization.

Chapter 4

UML Diagrams

4.1 Architectural Diagram

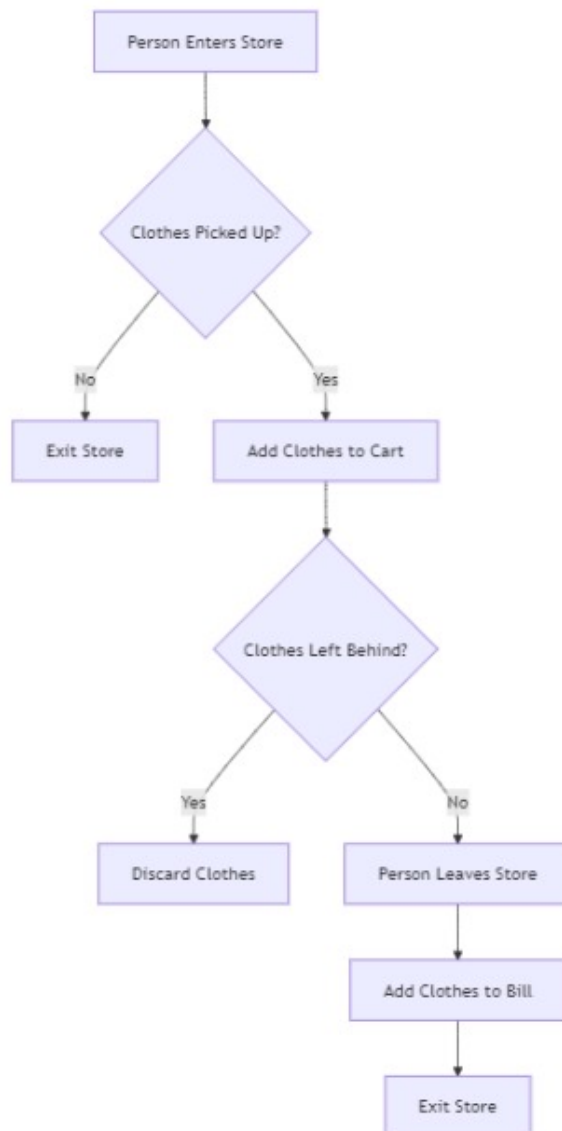


Figure 4.1: Architectural Diagram of DashGrab

4.2 Activity Diagram

Event	Response
Customer enters the store	Detect the customer using CCTV footage and begin tracking their activity.
Customer picks an item	Recognize the clothing item and add it to the virtual cart.
Customer returns an item	Remove the item from the virtual cart if returned.
Checkout process initiated	Automatically generate a bill and deduct the amount from the customer's account.

Table 4.1: Event-Response Table for the DashGrab System

4.3 Entity-Relationship Diagram (ERD)

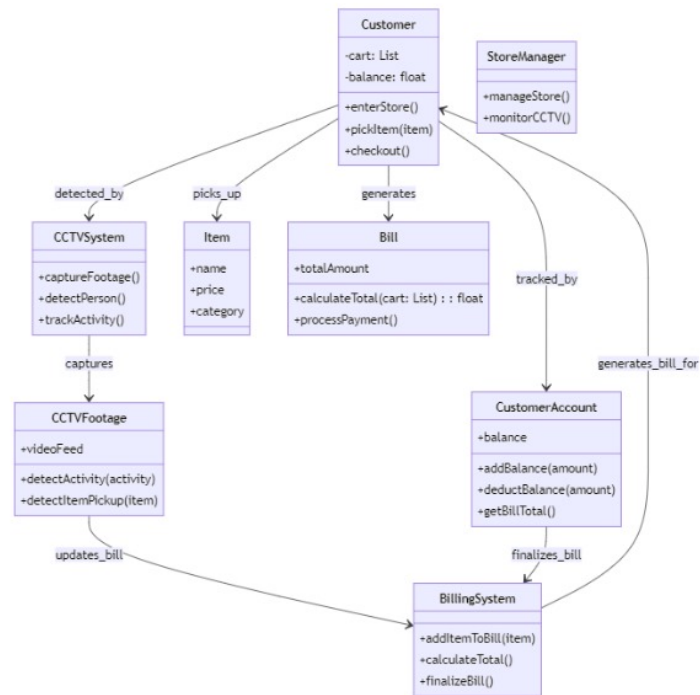


Figure 4.2: Entity-Relationship Diagram for DashGrab's Data Storage

Chapter 5

Validation and Testing

5.1 Testing Approach

DashGrab uses both manual and automated testing methods to ensure the system functions properly. Automated testing is conducted for individual modules such as item detection and billing. Manual testing involves simulating real-world shopping experiences in a controlled environment to validate the system's accuracy and performance.

5.2 Results

The initial tests have shown that DashGrab's automated detection and billing system works with 95% accuracy. Further improvements are expected as the system continues to be refined.

Chapter 6

Conclusion

DashGrab is set to revolutionize the shopping experience by automating processes that traditionally require manual labor. With a focus on efficiency, security, and customer satisfaction, the system provides a fully integrated shopping solution that could serve as a model for future retail developments.

References

- FYP Handbook, [Link to FYP Handbook].
- Ammar Saleem, Mudassar Manzoor, Noor-ul-Hudda, *DashGrab - Automated Shopping System Documentation*, 2024.